

Open Source Workshop
“Spin The Wheels” Interaction
Specification Brief
Curious Learning

Product Specification for
“Creating Literacy Games with React Native” workshop

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Glossary

MVP

Minimum Viable Product. A small, testable piece of completed software that provides value to a user.

Project Difficulty

This project's difficulty rating is: Beginner / Intermediate

This project does not have an existing code base for reference.

This project must be completed using English-language content only.

Purpose/objective

The pedagogical objective of the “Spin The Wheels” interaction paradigm is to put order to a set of letters to spell a word. An example is having 3 independently spinning wheels, each with a set of letters, viewed from a fixed perspective. As you turn the wheels, the letters could form a word or could not form a word-- similar to a combination letter lock.

MVP Implementation

UI Requirements

- An area with three slots, or spinners, of letters. Each spinner will have a predefined set of letters that can go in that particular slot.
- A mechanism for randomly choosing a word to spell from a JSON list and have the letters appear in the correct slots.
- An area to display a picture of the spelled word.

Functional Requirements

- Upon interacting with the mechanism for randomly choosing a word to spell, the UI will animate the spinners spinning between letters before slowing down to settle on a particular letter. The letters must spell a word.
- When the letters displayed spell a word, an image of the word should be displayed in the picture area and an audio file should play pronouncing the word.

Constraints

- The middle slot letter can only be a vowel.
- All content must be English-only.

Input Requirements

- Your program must be able to read in a JSON file containing information about the spinners and containing a list of words they can spell.

Provided Inputs

You will be given a JSON file which will define the spinners, what letters appear on each spinner, and a word list that the spinners can spell.

The basic structure of the JSON will look something like:

```

{
  // An array of all spinners
  "spinners" : [
    "spinner1" : {
      "letter_list" : char[ ] // all letters that can go on a particular spinner
    },
    ...
    "spinnerN" : {
      "letter_list" : char[ ] // all letters that can go on a particular spinner
    }
  ],
  "level_id" : string, // a unique ID to identify the level
  "word_list" : string[ ] // an array of all the words the spinners could spell
}

```

If the JSON file defines only 3 spinners, then the second spinner will contain only vowels,

Media Asset Considerations

In this app, an object will also have an association with:

- An audio file pronouncing the word.
- An image of the word.

Better Implementation

Additional UI Requirements

- [A] mechanism(s) for moving one letter at a time for each of the slots.

Additional Functional Requirements

- If the letters chosen do not display a real word, a placeholder character graphic showing confusion with an audio file of gibberish being pronounced will be played.
- If the child has completed 50% of the word list (either through changing the spinners or through pulling the lever), move on to a new word list and new spinners.

Great Implementation

Additional Functional Requirements

- Add the ability to flick to “spin” each spinner manually. The velocity of the flick should determine how fast the spinner spins.
- Add support for spelling words greater than 3 letters.